SG No. 53 D_{2h}^{7} $Pmna$ [orthorhombic]

* plus set: $+[0,\ 0,\ 0]$

Table 1: Wyckoff site: `2a`, site symmetry: `2/m..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,4,5,8]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,3,6,7]`

Table 2: Wyckoff site: `2b`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,4,5,8]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[2,3,6,7]`

Table 3: Wyckoff site: `2c`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[1,4,5,8]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,3,6,7]`

Table 4: Wyckoff site: `2d`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,4,5,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,3,6,7]`

Table 5: Wyckoff site: `4e`, site symmetry: `2..`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,4]`
2	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	`[2,3]`
3	$[-x,\ 0,\ 0]$	`[5,8]`
4	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[6,7]`

Table 6: Wyckoff site: `4f`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,4]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,3]`
3	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[5,8]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[6,7]`

Table 7: Wyckoff site: `4g`, site symmetry: `.2.`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ \frac{1}{4}\right]$	`[1,3]`
2	$\left[\frac{1}{4},\ -y,\ \frac{3}{4}\right]$	`[2,4]`
3	$\left[\frac{3}{4},\ -y,\ \frac{3}{4}\right]$	`[5,7]`
4	$\left[\frac{3}{4},\ y,\ \frac{1}{4}\right]$	`[6,8]`

Table 8: Wyckoff site: `4h`, site symmetry: `m..`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,8]`
2	$\left[\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	`[2,7]`
3	$\left[\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	`[3,6]`
4	$[0,\ -y,\ -z]$	`[4,5]`

Table 9: Wyckoff site: `8i`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	`[2]`
3	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	`[3]`
4	$[x,\ -y,\ -z]$	`[4]`
5	$[-x,\ -y,\ -z]$	`[5]`
6	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	`[6]`
7	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	`[7]`
8	$[-x,\ y,\ z]$	`[8]`